5.9.1

EE25BTECH11047 - RAVULA SHASHANK REDDY

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Question:

Consider the basis

$$\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\} \text{ of } \mathbb{R}^3, \text{ where } \mathbf{u}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \ \mathbf{u}_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \ \mathbf{u}_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

Let $\{\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3\}$ be the dual basis and $f(a, b, c) = a + b + c$. If

$$\mathbf{f} = \alpha_1 \mathbf{f}_1 + \alpha_2 \mathbf{f}_2 + \alpha_3 \mathbf{f}_3,$$

find $\begin{pmatrix} \alpha_1 & \alpha_2 & \alpha_3 \end{pmatrix}$.

Solution:

The dual basis satisfies :

$$\mathbf{f}_i(\mathbf{u}_j) = \delta_{ij} = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}. \tag{1}$$

$$\mathbf{U} = \begin{pmatrix} \mathbf{u}_1 \ \mathbf{u}_2 \ \mathbf{u}_3 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \tag{2}$$

$$\mathbf{F} = \begin{pmatrix} \mathbf{f}_1 \\ \mathbf{f}_2 \\ \mathbf{f}_3 \end{pmatrix}. \tag{3}$$

Then the dual basis condition becomes

$$\mathbf{F}\,\mathbf{U} = \mathbf{I_3}. \tag{4}$$

$$\mathbf{F}^{-1} = \mathbf{U} \tag{5}$$

Given:

$$f\left(a,b,c\right) = a + b + c = \begin{pmatrix} 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} \tag{6}$$

$$f(\mathbf{x}) = \mathbf{f}\mathbf{x} \tag{7}$$

$$\text{where} \quad \mathbf{f} = \begin{pmatrix} 1 & 1 & 1 \end{pmatrix} \tag{8}$$

Also given that

$$\mathbf{f} = \alpha_1 \mathbf{f}_1 + \alpha_2 \mathbf{f}_2 + \alpha_3 \mathbf{f}_3 \implies \mathbf{f} = \mathbf{a}^T \mathbf{F}, \tag{9}$$

$$\text{where} \quad \mathbf{a} = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} \tag{10}$$

$$\mathbf{a}^T \mathbf{F} = \mathbf{f} \implies \mathbf{a}^T = \mathbf{f}\mathbf{F}^{-1} = \mathbf{f}\mathbf{U}, \tag{11}$$

$$\mathbf{a}^T = \begin{pmatrix} 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \end{pmatrix}. \tag{12}$$

$$\begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \tag{13}$$